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In [ ]: Name :-Mayuri Hirade
Roll no. :- 13163
Batch :- A3
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```
In [16]: pip install tabulate
```

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: tabulate in c:\programdata\anaconda3\lib\site-packages (0.9.0)
Note: you may need to restart the kernel to use updated packages.

```
In [20]: import pandas as pd
```

```
dataset = pd.read_csv("Social_Network_Ads.csv")

from tabulate import tabulate
print(tabulate(dataset.head(), headers='keys', tablefmt='grid'))
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
In [18]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

def sigmoid(z):
    return 1 / (1 + np.exp(-z))

X_linear = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
y_linear = np.array([2, 4, 6, 8, 10])

linear_model = LinearRegression()
linear_model.fit(X_linear, y_linear)
y_pred_linear = linear_model.predict(X_linear)

X_logistic = np.linspace(-6, 6, 100)
y_logistic = sigmoid(X_logistic)

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.scatter(X_linear, y_linear, color='blue', label="Actual Data")
plt.plot(X_linear, y_pred_linear, color='red', label="Linear Regression")
plt.title("Linear Regression: y = b0 + b1*x")
```

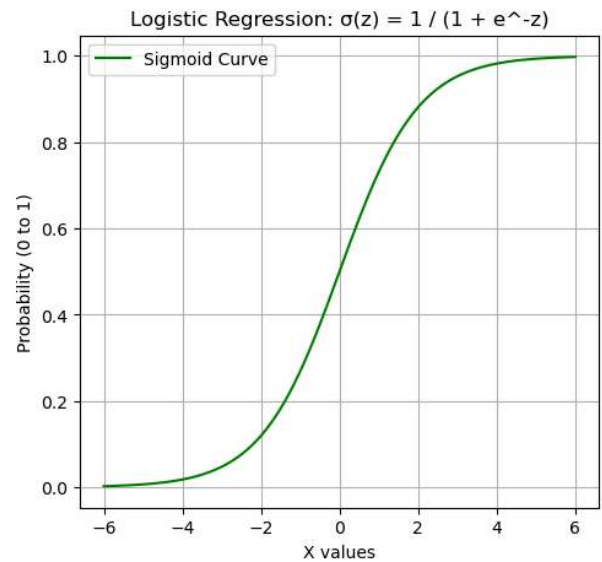
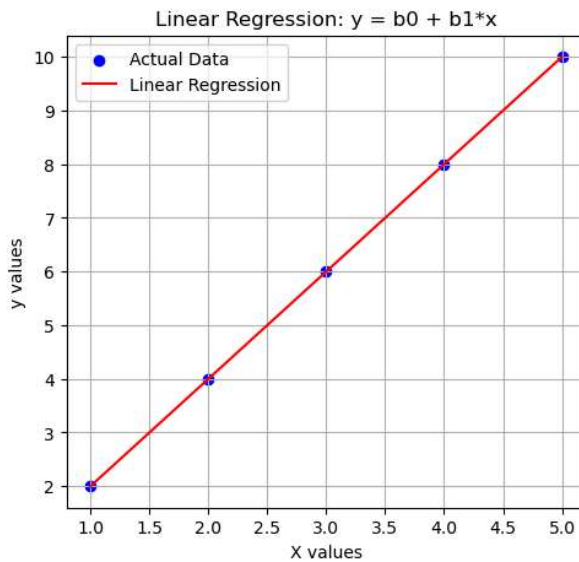
```

plt.xlabel("X values")
plt.ylabel("y values")
plt.legend()
plt.grid()

plt.subplot(1, 2, 2)
plt.plot(X_logistic, y_logistic, color='green', label="Sigmoid Curve")
plt.title("Logistic Regression:  $\sigma(z) = 1 / (1 + e^{-z})$ ")
plt.xlabel("X values")
plt.ylabel("Probability (0 to 1)")
plt.legend()
plt.grid()

plt.show()

```



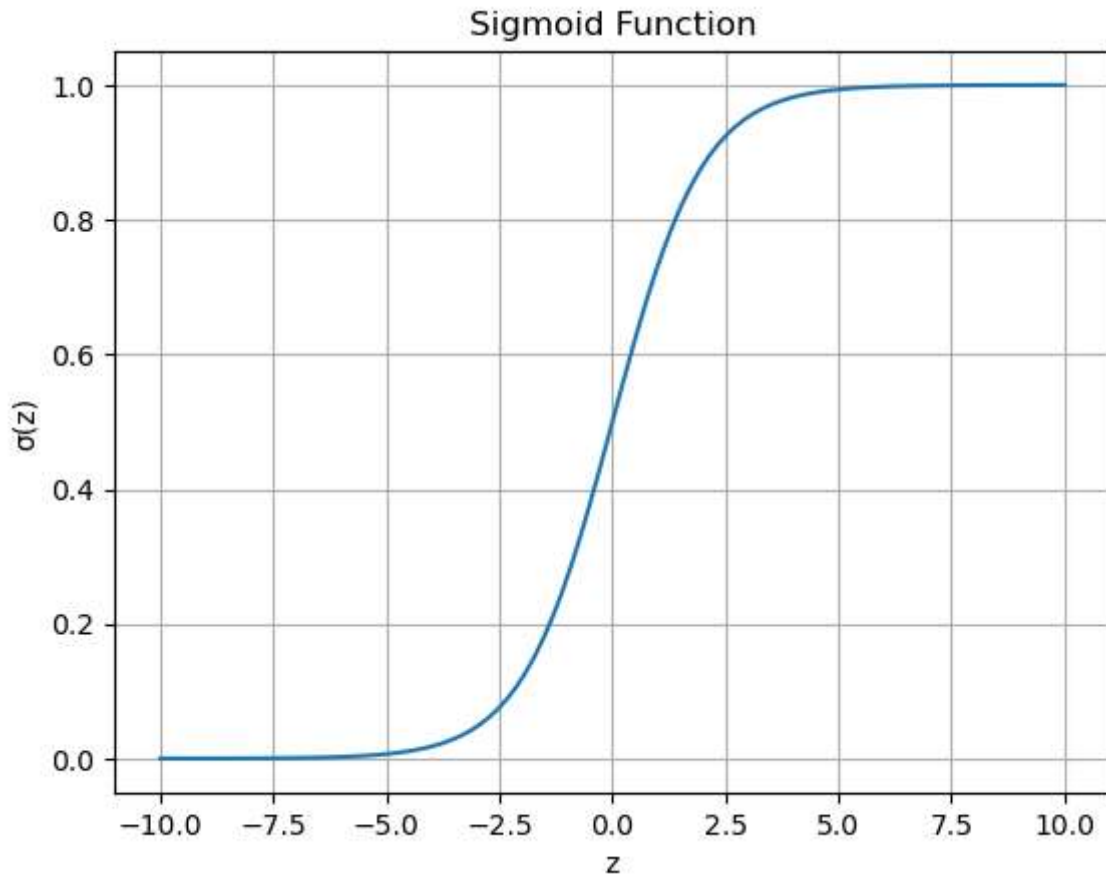
```

In [19]: import numpy as np
import matplotlib.pyplot as plt

def sigmoid(z):
    return 1 / (1 + np.exp(-z))

z = np.linspace(-10, 10, 100)
plt.plot(z, sigmoid(z))
plt.title("Sigmoid Function")
plt.xlabel("z")
plt.ylabel(" $\sigma(z)$ ")
plt.grid()
plt.show()

```



```
In [24]: import pandas as pd
import numpy as np
import warnings
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, classification_report
from tabulate import tabulate

warnings.simplefilter(action='ignore', category=FutureWarning)

dataset = pd.read_csv("Social_Network_Ads.csv")

print("\nDataset Preview:\n")
print(tabulate(dataset.head(), headers='keys', tablefmt='grid'))

X = dataset[['Age', 'EstimatedSalary']].values
y = dataset['Purchased'].values

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_st

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

multi_logistic = LogisticRegression(solver='lbfgs')
multi_logistic.fit(X_train, y_train)
```

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y_pred_multi = multi_logistic.predict(X_test)
print("\nLogistic Regression Predictions:\n", y_pred_multi)

cm = confusion_matrix(y_test, y_pred_multi)
cm_df = pd.DataFrame(cm, index=["Actual 0", "Actual 1"], columns=["Predicted 0", "P
print("\nConfusion Matrix:\n")
print(tabulate(cm_df, headers='keys', tablefmt='grid'))

report_df = pd.DataFrame(classification_report(y_test, y_pred_multi, output_dict=Tr
print("\nClassification Report:\n")
print(tabulate(report_df, headers='keys', tablefmt='grid'))
```

Dataset Preview:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
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Logistic Regression Predictions:

```
[0 1 0 1 0 0 1 0 0 0 0 1 0 0 0 0 1 1 0 1 0 0 0 1 0 0 1 0 1 0 0 0 1 0 1 0 0
0 0 0 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
0 0 1 1 0 0 0 0 1 0 1 0 0 1 0 0 1 0 0 0 0 0 1 0 1 0]
```

Confusion Matrix:

	Predicted 0	Predicted 1
Actual 0	61	2
Actual 1	12	25

Classification Report:

	precision	recall	f1-score	support
0	0.835616	0.968254	0.897059	63
1	0.925926	0.675676	0.78125	37
accuracy	0.86	0.86	0.86	0.86
macro avg	0.880771	0.821965	0.839154	100
weighted avg	0.869031	0.86	0.85421	100

```
In [25]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix
```

```
warnings.simplefilter(action='ignore', category=FutureWarning)

dataset = pd.read_csv("Social_Network_Ads.csv")

X = dataset[['Age', 'EstimatedSalary']].values
y = dataset['Purchased'].values

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_st

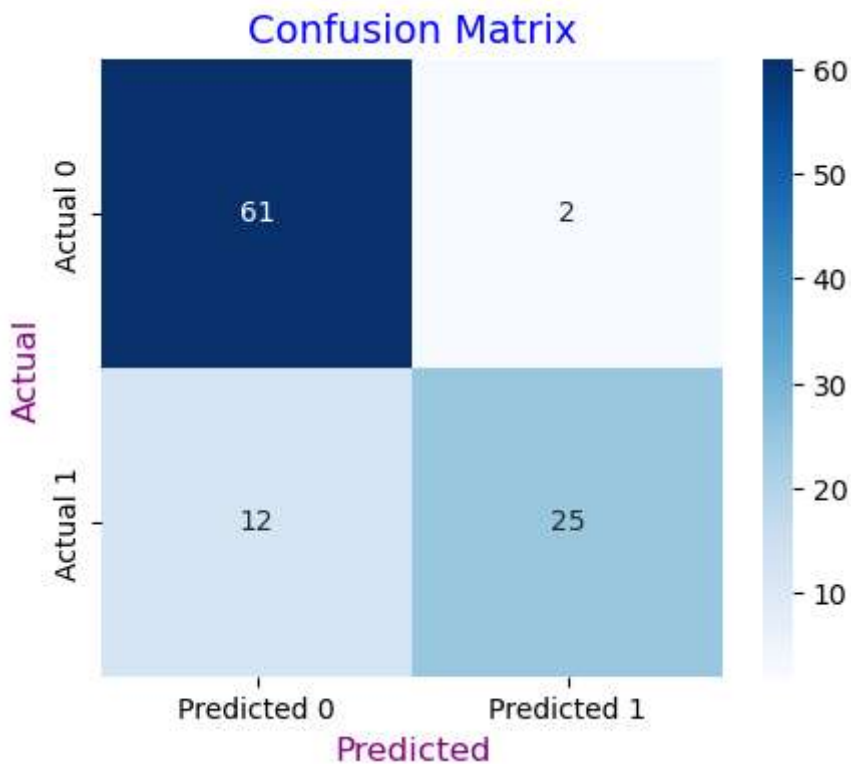
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

logistic_model = LogisticRegression(solver='lbfgs')
logistic_model.fit(X_train, y_train)

y_pred = logistic_model.predict(X_test)

cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(5, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=["Predicted 0", "Pre
plt.xlabel("Predicted", fontsize=12, color="purple")
plt.ylabel("Actual", fontsize=12, color="purple")
plt.title("Confusion Matrix", fontsize=14, color="blue")
plt.show()
```



In []: